



Sir Syed University
of Engineering & Technology

Operating System Lab

Project Report

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Operating Systems Lab Project

BS Computer Science, Batch 2022F

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Project Report

❖ Scheduling Algorithm

1. Introduction

The First Come First Serve (FCFS) and Round Robin (RR) scheduling algorithms are two fundamental CPU scheduling techniques widely studied and implemented in operating systems. This project aims to provide a comprehensive understanding and practical implementation of both algorithms using a C# Windows Form GUI and SQL Server for data management. The application offers an intuitive and user-friendly interface, allowing users to interact seamlessly with the scheduling algorithms. By incorporating both free and premium features, the application caters to a wide range of users, enhancing their learning experience and providing valuable insights into CPU scheduling.

2. Project Objective

The primary objective of this project is to create a user-friendly application that demonstrates the functionality of both the FCFS and RR scheduling algorithms. The application should allow users to:

- Register and log in with unique credentials.
- Input and manage process data.
- Calculate and display key scheduling metrics such as completion time, turnaround time, and waiting time.
- Provide additional premium features such as Gantt Chart visualization and detailed breakdowns of each calculation step for paid users.
- Collect user feedback for further improvements.

3. Features

The project includes the following features:

a) User Authentication

- **Login:** Users log in by entering their username and password.
- **Registration:** New users can create an account by providing their name, a unique username, and a password.

b) Home Page

- **Poster Display:** A visual representation or poster of the FCFS and RR algorithms is displayed on the home page.

c) Process Management

- **Add Process:** Users can add process details (arrival time, burst time, process ID) which are displayed in a DataGridView.
- **Choose Algorithm:** Users can select either FCFS or RR scheduling algorithms.
- **Input Time Quantum:** Users can input the time quantum for the RR scheduling algorithm.
- **Calculate Metrics:** Calculates and displays completion time, turnaround time, and waiting time for each process, along with average turnaround time and average waiting time.
- **Clear Processes:** Allows users to clear the entered process data from the DataGridView.

d) Premium Features

- **Upgrade to Pro:** Users can upgrade to the pro version by paying \$9 to access additional features.
- **Pro Features:** Includes Gantt Chart visualization and detailed breakdown of calculation steps.
- **Payment Integration:** Users can enter card details (Union, Visa, VisaMaster, Mastercard) to complete the upgrade.

e) Feedback

- **User Feedback:** Users can provide feedback about the application.

4. Problem Statements

- **User Authentication:** Ensuring secure and unique user registration and login.
- **Data Management:** Efficiently storing and retrieving user and process data using SQL Server.
- **Calculation Accuracy:** Accurately calculating and displaying scheduling metrics for both FCFS and RR algorithms.
- **User Interface:** Designing an intuitive and responsive GUI for a seamless user experience.
- **Payment Processing:** Securely handling payment transactions for upgrading users to the pro version.
- **Feature Access Control:** Differentiating between free and pro users to restrict or grant access to premium features

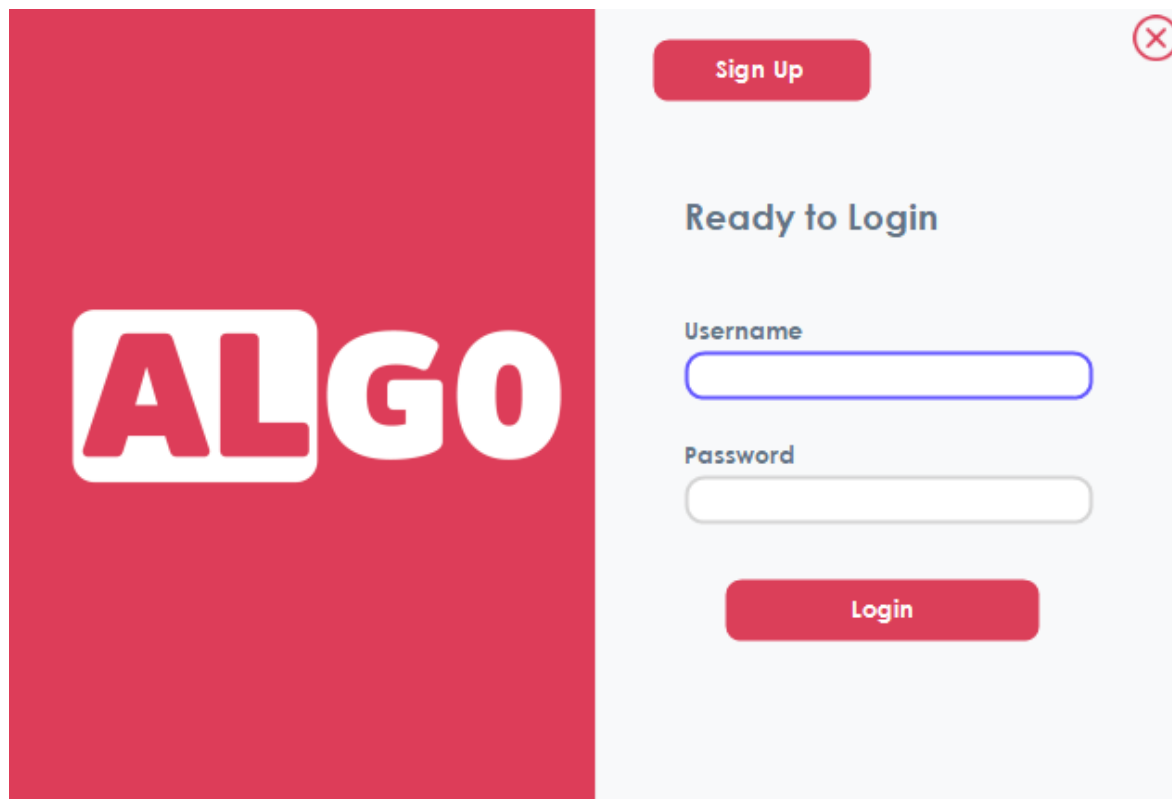
5. Technologies Used

- **Programming Language:** C#
- **Framework:** .NET Framework for Windows Forms
- **Database:** SQL Server for storing user credentials and process data
- **GUI Design:** Windows Forms for creating the user interface

6. Conclusion

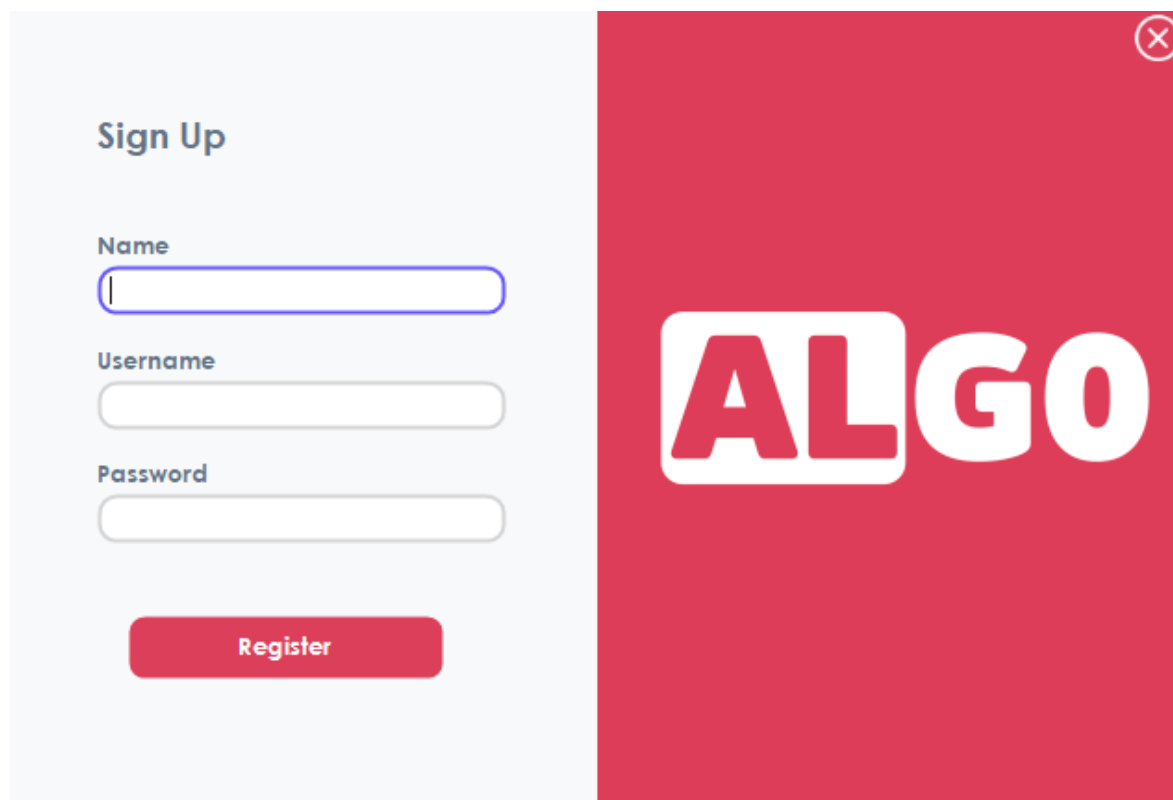
The Scheduling Algorithm project successfully demonstrates the implementation of both the FCFS and RR algorithms using a C# Windows Form application with SQL Server integration. The project achieves its objectives by providing essential scheduling functionalities along with a user-friendly interface. The additional premium features offer enhanced visualization and detailed breakdowns, making the application both educational and practical for users. Future enhancements could include adding more scheduling algorithms and further improving the user interface based on feedback.

Upon launching the application, the user is presented with the login form.



The image shows a login interface for 'ALGO'. On the left is a red vertical panel with the 'ALGO' logo in white. On the right is a light gray panel with a 'Sign Up' button at the top. Below it is the heading 'Ready to Login'. There are two input fields: 'Username' and 'Password'. At the bottom of the gray panel is a red 'Login' button. A red close button (X) is in the top right corner of the gray panel.

If the user has not already signed up, they can do so before logging in. The username must be unique. If the username is already taken, the application will display an error message.



The image shows a sign up interface for 'ALGO'. On the left is a light gray panel with the heading 'Sign Up'. Below it are three input fields: 'Name', 'Username', and 'Password'. At the bottom of the gray panel is a red 'Register' button. On the right is a red vertical panel with the 'ALGO' logo in white. A red close button (X) is in the top right corner of the red panel.

Once the user signs in, the home page is displayed, showcasing a poster for FCFS and Round Robin scheduling algorithms

ALGO

Welcome, Faisal

Logout

Home

Calculate

Upgrade

Feedback

First Come First Serve

Round Robin

Scheduling Algo

Operating System

FCFS (Example)

Process	Duration	Order	Arrival Time
P1	24	1	0
P2	3	2	0
P3	4	3	0

Gantt Chart:

P1(24)P2(3)P3(4)

When the user clicks the "Calculate" button, they can choose between FCFS (default) and Round Robin scheduling algorithms. They can add or clear processes. The application calculates and displays the waiting time, completion time, and turnaround time in a DataGridView, along with the average TAT and WT

ALGO

Welcome, Faisal

Logout

Home

Calculate

Upgrade

Feedback

Process ID

Arrival Time

Burst Time

1

0

0

Add Process

Clear Process

FCFS

Round Robin

Calculate

First come first serve

Process Id	Arrival Time	Burst Time	Complete Time	TurnAround Time	Waiting Time

Average Time

Average TurnAround Time:

Average Waiting Time:

Show Steps

Welcome, Faisal

Logout

Home

Calculate

Upgrade

Feedback

Process ID

1

Arrival Time

0

Burst Time

0

Add Process

FCFS

Round Robin

Time Quantum

1

Calculate

Clear Process

Round Robin

Process Id	Arrival Time	Burst Time	Complete Time	TurnAround Time	Waiting Time

Average Time

Average TurnAround Time:

Average Waiting Time:

Show Steps

When the user clicks "Show Steps," if they have not upgraded to Pro, the application displays an error message instructing them to upgrade to Pro first

Welcome, Faisal

Logout

Home

Calculate

Upgrade

Feedback

Process ID

3

Arrival Time

2

Burst Time

5

Add Process

FCFS

Round Robin

First come first serve

Process Id	Arrival Time	Burst Time	Complete Time	TurnAround Time	Waiting Time
1	0	2	2	2	
2	1	4	6	5	
3	2	5	11	9	4

Upgrade to pro

OK

Average Time

Average TurnAround Time:

5.33

Average Waiting Time:

1.67

Show Steps

When the user clicks the upgrade button, a panel appears showing the benefits of the Pro version. The user can then upgrade by clicking the Pro button.



Welcome, Faisal

Logout

 Home Calculate Upgrade Feedback

Free

-  **Calculation**
Users input processes, and the app calculates completion, turnaround, and waiting times
-  **No Steps Explanation**
Direct calculation results are displayed without a detailed explanation of each step.
-  **No Gantt Chart**
The free version does not include a visual Gantt chart representation of the process schedule

Pro

-  **Calculation**
Users input processes, and the app calculates completion, turnaround, and waiting times
-  **Step-by-Step Explanation**
Detailed breakdown of each calculation step, providing transparency and educational value.
-  **Gantt Chart Visualization**
Visual representation of the FCFS scheduling using a Gantt chart for better understanding.

Upgrade to pro

When the user clicks the "Upgrade to Pro" button, a new form appears where they can input their card number, CVV, card expiry date. The upgrade costs \$9 for a lifetime subscription. Only UnionPay, Visa, Visa Master, and MasterCard are accepted.



Upgrade to pro

Card Number

CVV

Card Expiry Date

Lifetime

\$9

Pay



If the user has already paid and clicks the "Upgrade to Pro" button again, an error message will be displayed indicating that they have already paid.

ALGO

Welcome, Faisal

Logout

Home

Calculate

Upgrade

Feedback

Free

✓ Calculation

Users input processes, and the app calculates completion, turnaround, and waiting times

✓ No Steps Explanation

Direct calculation results are displayed with a detailed explanation of each step.

✓ No Gantt Chart

The free version does not include a visual Gantt chart representation of the process schedule

Pro

✓ Calculation

Users input processes, and the app calculates completion, turnaround, and waiting times

✓ Step-by-Step Explanation

Detailed breakdown of each calculation step, providing transparency and educational value.

✓ Gantt Chart Visualization

Visual representation of the FCFS scheduling using a Gantt chart for better understanding.

Upgrade to pro

Error Message

✗

You have already paid

OK

First come First Serve (FCFS) example:

ALGO

Welcome, Faisal

Logout

Home

Calculate

Upgrade

Feedback

Process ID

Arrival Time

Burst Time

FCFS

Round Robin

Add Process

Clear Process

5

4

5

Calculate

First come first serve

Process Id	Arrival Time	Burst Time	Complete Time	TurnAround Time	Waiting Time
1	0	4	4	4	0
2	1	3	7	6	3
3	2	1	8	6	5
4	3	2	10	7	5
5	4	5	15	11	6

Show Steps

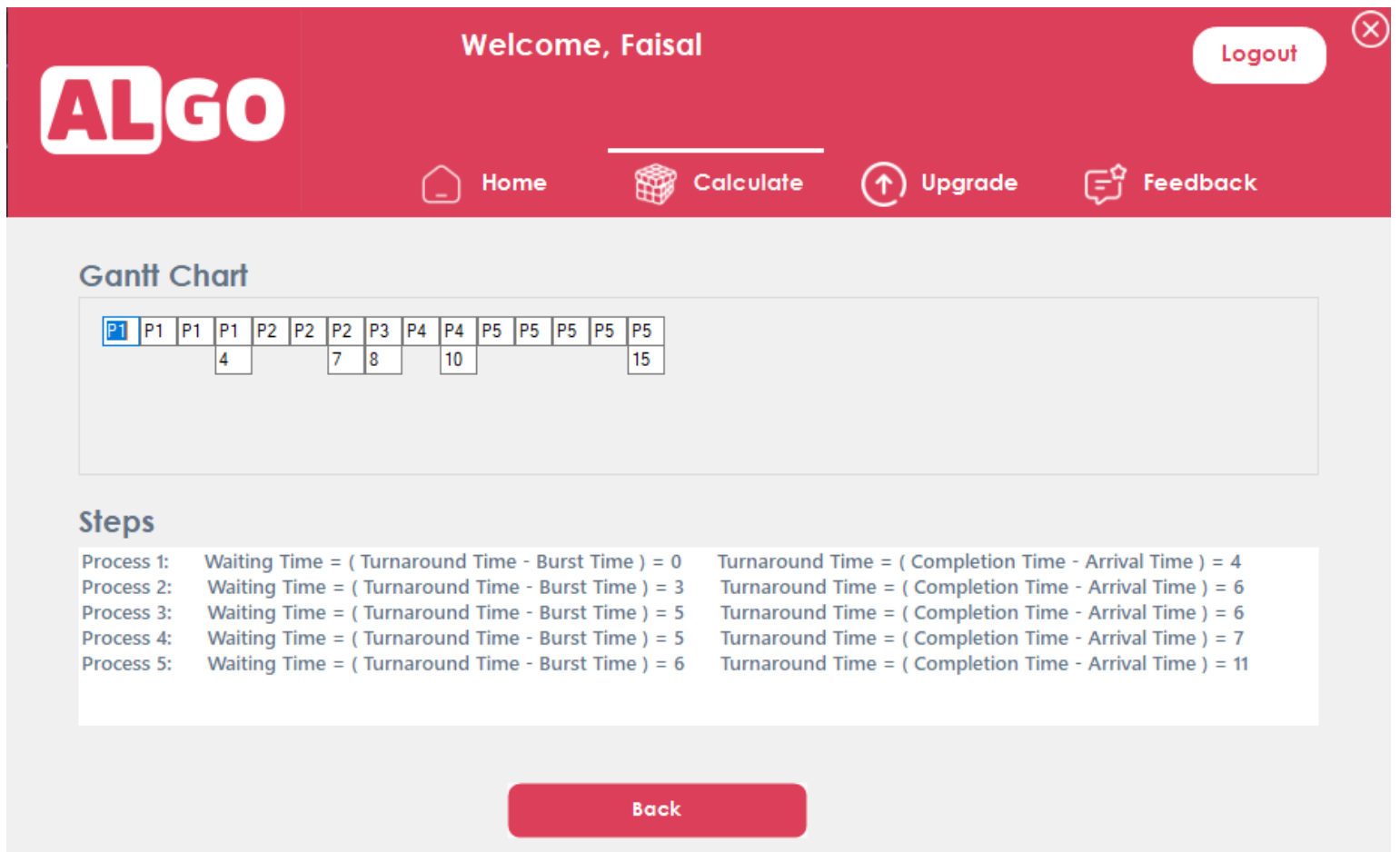
Average Time

Average TurnAround Time:

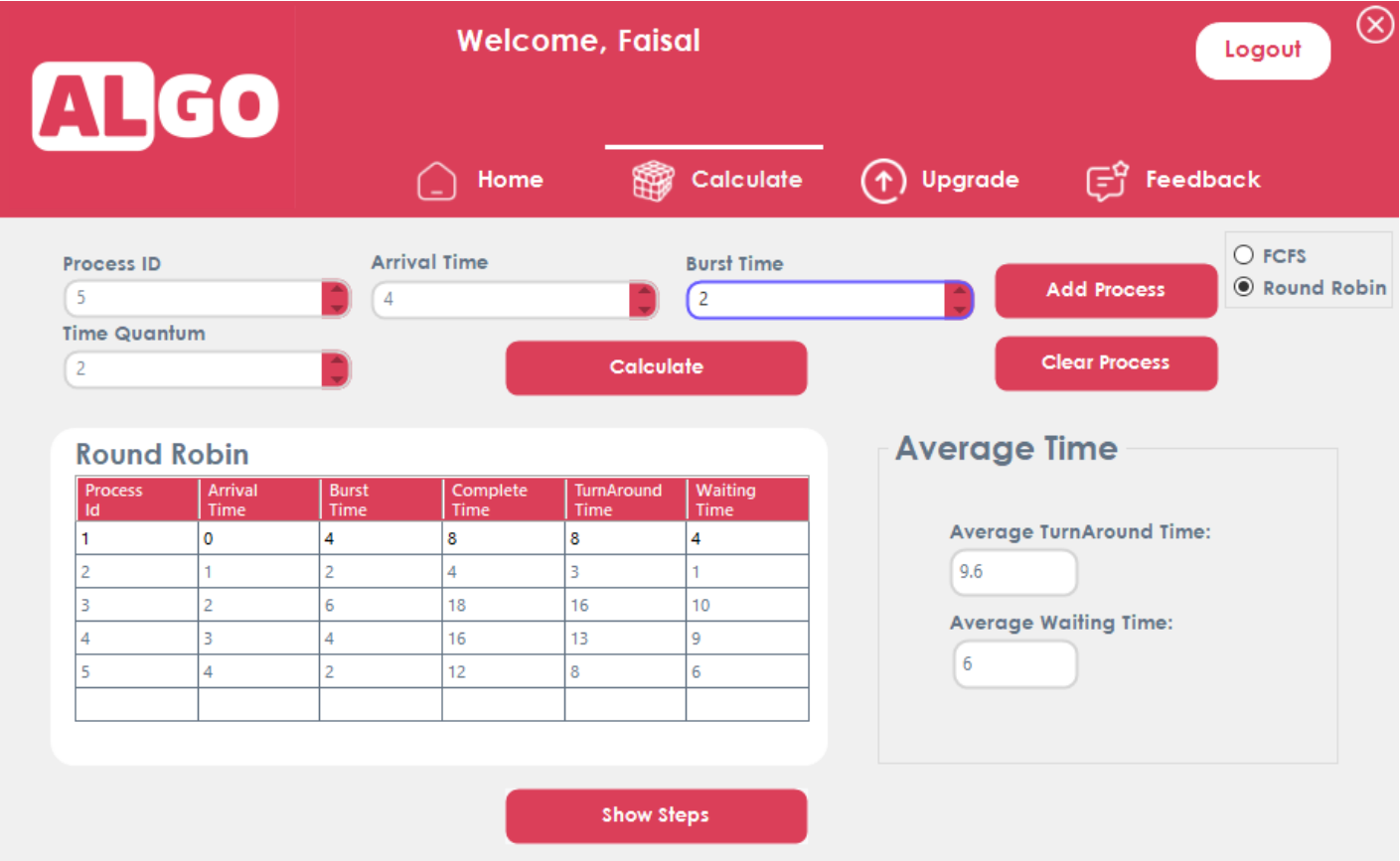
6.8

Average Waiting Time:

3.8



Round Robin (RR) example:



ALGO

Welcome, Faisal

Logout

Home

Calculate

Upgrade

Feedback

Gantt Chart

P1	P1	P2	P2	P3	P3	P1	P1	P4	P4	P5	P5	P3	P3	P4	P4	P3	P3
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	18

Steps

Process 1: Waiting Time = (Turnaround Time - Burst Time) = 4

Process 2: Waiting Time = (Turnaround Time - Burst Time) = 1

Process 3: Waiting Time = (Turnaround Time - Burst Time) = 10

Process 4: Waiting Time = (Turnaround Time - Burst Time) = 9

Process 5: Waiting Time = (Turnaround Time - Burst Time) = 6

Turnaround Time = (Completion Time - Arrival Time) = 8

Turnaround Time = (Completion Time - Arrival Time) = 3

Turnaround Time = (Completion Time - Arrival Time) = 16

Turnaround Time = (Completion Time - Arrival Time) = 13

Turnaround Time = (Completion Time - Arrival Time) = 8

Back

The user has the option to write feedback in the designated feedback section.

ALGO

Welcome, Faisal

Logout

Home

Calculate

Upgrade

Feedback

Feedback

Write feedback about us.

Submit

❖ Bash Script for Evaluating Arithmetic Expressions with a Custom Flag

1. Introduction

This project aims to develop a Bash script capable of evaluating arithmetic expressions both as separate arguments and as a single expression string. This script serves as a command-line tool for performing basic arithmetic operations such as addition, subtraction, multiplication, and division. It enhances the understanding of scripting, command-line tools, and basic arithmetic computation in the Linux environment..

2. Project Objective

The main objective of this project is to create a versatile and user-friendly Bash script that can:

- Accept arithmetic expressions as individual arguments.
- Evaluate complete arithmetic expressions provided as a single string.
- Provide clear usage instructions and examples.
- Handle different arithmetic operations accurately, including addition, subtraction, multiplication, and division.

3. Problem Statements

The project addresses the following problems:

- **Ease of Use:** Users need a simple and straightforward method to perform arithmetic calculations from the command line.
- **Flexibility:** The script should support both individual arguments for numbers and operators, as well as complete expressions provided as a single string.
- **Accuracy:** The script must handle different arithmetic operations accurately and return results with appropriate precision.
- **User Guidance:** The script should provide clear instructions and usage examples to help users understand how to use it effectively.

4. Technologies Used

The following technologies and tools were used in this project:

- **Bash Scripting:** The core of the project is written in Bash, a Unix shell and command language, enabling the creation of powerful and flexible command-line tools.
- **bc (Basic Calculator):** A command-line calculator that supports arbitrary precision arithmetic, used to evaluate expressions and perform arithmetic operations.
- **Linux Command Line:** The script is designed to run in a Linux environment, leveraging the power of the command line for user interaction and execution..

5. Command

- **Source Code**

```
faisal152@faisal152:~$ cd /usr/bin
```

```
faisal152@faisal152:/usr/bin$ sudo nano calc
```

```
faisal152@faisal152:/usr/bin$ sudo chmod 755 calc
```

```
#!/bin/bash
```

```
# Function to display usage
```

```
show_help() {  
    echo "Usage: $0 [options] <num1> <operator> <num2>"  
    echo "Options:"  
    echo "  -e, --expression    Provide arithmetic expression in a single string"  
    echo "  -h, --help          Show this help message and exit"  
    echo  
    echo "Examples:"  
    echo "  $0 5 + 3"  
    echo "  $0 5 '*' 3"  
    echo "  $0 5 \\* 3"  
    echo "  $0 -e \"(5 + 3) * 2\""  
    echo "  $0 -e \"2*(6+3)\""  
    echo "  $0 -e \"3+5*2-8/4\""  
}
```

```
# Function to evaluate an arithmetic expression
```

```
evaluate_expression() {  
    local expression=$1  
    result=$(echo "scale=2; $expression" | bc -l)  
    printf "Result: %.2f\\n" "$result"  
}
```

```
# Function to perform basic arithmetic operations
```

```
perform_arithmetic() {  
    local num1=$1  
    local operator=$2  
    local num2=$3  
  
    case $operator in  
        +) result=$(echo "scale=2; $num1 + $num2" | bc);;  
        -) result=$(echo "scale=2; $num1 - $num2" | bc);;  
        '*') result=$(echo "scale=2; $num1 * $num2" | bc);;  
        /) result=$(echo "scale=2; $num1 / $num2" | bc);;  
        *) echo "Invalid operator"; exit 1;;  
    esac  
  
    printf "Result: %.2f\\n" "$result"  
}
```

```
# Check for flags
```

```
while [[ $# -gt 0 ]]; do  
    case $1 in  
        -e|--expression)  
            expression_mode=true  
            expression=$2  
            shift 2  
            ;;  
        -h|--help)  
            show_help  
            exit 0  
            ;;  
        *)  
            break  
            ;;  
    esac  
done
```

```
# If expression mode is enabled, evaluate the expression
if [ "$expression_mode" = true ]; then
    evaluate_expression "$expression"
    exit 0
fi

# Check if the correct number of arguments is provided
if [ "$#" -ne 3 ]; then
    echo "Usage: $0 <num1> <operator> <num2>"
    exit 1
fi

# Perform the basic arithmetic operation
perform_arithmetic $1 "$2" $3
```

- File location

```
faisal152@faisal152:~$ which calc
/usr/bin/calc
```

- Command implementation images

```
faisal152@faisal152:/usr/bin$ calc 5 + 3
Result: 8.00
```

```
faisal152@faisal152:/usr/bin$ calc 5 - 2
Result: 3.00
```

```
faisal152@faisal152:/usr/bin$ calc 5 / 3
Result: 1.66
```

```
faisal152@faisal152:/usr/bin$ calc 5 \* 2
Result: 10.00
```

```
faisal152@faisal152:/usr/bin$ calc 5 '*' 3
Result: 15.00
```

6. Flag

- -e or --expression Flag

Purpose:

This flag allows the user to provide an entire arithmetic expression as a single string. It signals the script to evaluate the given expression directly, rather than interpreting separate arguments for numbers and operators.

Flag implementation images:

```
faisal152@faisal152:/usr/bin$ calc --expression "4+3/2*(3+7)"
Result: 19.00
```

```
faisal152@faisal152:/usr/bin$ calc -e "(9 + 2) * 3 + 4 * 5"
Result: 53.00
```

- --h or --help Flag

Purpose:

This flag provides the user with a help message, explaining how to use the script and giving examples of valid commands. It is useful for guiding users on the correct usage of the script.

Flag implementation images:

```
faisal152@faisal152:/usr/bin$ calc -h
Usage: /usr/bin/calc [options] <num1> <operator> <num2>
Options:
  -e, --expression    Provide arithmetic expression in a single string
  -h, --help          Show this help message and exit

Examples:
  /usr/bin/calc 5 + 3
  /usr/bin/calc 5 '*' 3
  /usr/bin/calc 5 \* 3
  /usr/bin/calc -e "(5 + 3) * 2"
  /usr/bin/calc -e "2*(6+3)"
  /usr/bin/calc -e "3+5*2-8/4"
```

```
faisal152@faisal152:/usr/bin$ calc --help
Usage: /usr/bin/calc [options] <num1> <operator> <num2>
Options:
  -e, --expression    Provide arithmetic expression in a single string
  -h, --help          Show this help message and exit

Examples:
  /usr/bin/calc 5 + 3
  /usr/bin/calc 5 '*' 3
  /usr/bin/calc 5 \* 3
  /usr/bin/calc -e "(5 + 3) * 2"
  /usr/bin/calc -e "2*(6+3)"
  /usr/bin/calc -e "3+5*2-8/4"
```